
1. INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Note Sharing Pro is an innovative web-based platform designed to revolutionize the note-sharing experience. The project aims to provide users with a seamless and convenient way to share various types of documents, PDFs, and more. This platform addresses the growing need for accessible and organized educational resources, facilitating a smoother exchange of knowledge and expertise.

The motivation behind Note Sharing Pro stems from the increasing demand for a centralized system where users can share and access high-quality educational materials. Traditional methods of sharing notes and resources, such as physical copies or scattered digital files, are often inefficient and difficult to manage. Note Sharing Pro provides a structured and user-friendly environment that not only simplifies document sharing but also incorporates a unique wallet system. This system allows users to earn virtual coins to access content, ensuring that contributors are rewarded for their efforts and users have access to valuable resources.

Moreover, the platform's global doubt section enhances its utility by allowing users to ask questions, seek clarifications, and engage in discussions with peers and experts. This feature promotes an interactive learning experience, making Note Sharing Pro a comprehensive tool for both individual study and collaborative learning.

1.2 THE PROPOSED SYSTEM

The Note Sharing Pro project connects users who need educational materials with those who can provide them. The system includes two primary stakeholders: the admin, the college and the users. The admin is responsible for managing site content, verifying colleges, monitoring user activity, and overseeing the virtual coin system. The college is responsible for managing faculty data, verifying documents, also monitoring user activity and overseeing the virtual coin system. Users, on the other hand, can register, upload documents, participate in the global doubt section, and utilize virtual coins to access content.

The platform offers several features designed to streamline the note-sharing process. Users can upload various document types, including question paper, hand written notes and PDFs, which are then verified by the college to ensure quality and relevance. The wallet system allocates free coins to new users, and additional coins can be purchased to view premium content. The global doubt section allows users to post questions, provide answers, and engage in meaningful discussions, fostering a collaborative learning environment.

By integrating these features, Note Sharing Pro makes it easier for users to find and share educational resources, enhancing the overall learning experience and promoting knowledge exchange on a global scale.

1.3 PROJECT SCOPE

1.3.1 Limitations of Existing System

1. Scattered and unorganized document sharing.
2. Lack of a reward system for contributors.
3. Limited interaction and discussion opportunities.
4. Inefficient verification and quality control of shared documents.
5. Difficulty in accessing diverse types of educational materials.

1.3.2 Advantages of Proposed System

1. Centralized platform for organized document sharing.
2. Reward system through virtual coins to incentivize contributors.
3. Interactive global doubt section for questions and discussions.
4. Admin verification to ensure document quality and relevance.
5. Easy access to various document types, question paper and PDFs.
6. Personalized user dashboards and efficient content management.

2. SYSTEM ANALYSIS

2.1 INTRODUCTION

Software Engineering is the analysis, design, construction, verification and management of technical or social entities. To engineer software accurately, a software engineering process must be defined. System analysis is a detailed study of the various operations performed by the system and their relationship within and module of the system. It is a structured method for solving the problems related to the development of a new system. The detailed investigation of the present system is the focal point of system analysis. This phase involves the study of parent system and identification of system objectives. Information has to be collected from all people who are affected by or who use the system. During analysis, data are collected on the variable files, decision point and transactions handled by the present system. The main aim of system is to provide the efficient and user-friendly automation. So, the system analysis process should be performed with extreme precision, so that an accurate picture of existing system, its disadvantages and the requirements of the new system can be obtained.

System analysis involves gathering the necessary information and using the structured tool for analysis. This includes the studying existing system and its drawback, designing a new system and conducting cost benefit analysis. System analysis is a problem-solving activity that requires intensive communication between the system users and system developers. The system is studied to the minute detail and analyzed. The system is viewed as a whole and the inputs to the system are identified. The outputs from the organization are traced through various phases of processing of inputs.

There are a number of different approaches to system analysis. When a computer-based information system is developed, systems analysis (according to the Waterfall model) would constitute the following steps:

- The development of a feasibility study, involving determining whether a project is economically, technologically and operationally feasible.
- Conducting fact-finding measures, designed to ascertain the requirements of the system's end-users. These typically span interviews, questionnaires, or visual observations of work on the existing system

- Gauging how the end-users would operate the system (in terms of general experience in using computer hardware or software), what the system would be used for and so on.

Techniques such as interviews, questionnaires etc. can be used for the detailed study of these processes. The data collected by these sources must be scrutinized to arrive at a conclusion.

The conclusion is an understanding of how the system functions. This system is called the Existing System. The Existing system is then subjected to close observation and the problem areas are identified. The designer now functions as a problem solver and tries to sort out the difficulties that the enterprise faces. The solutions are given as a proposal which is the Proposed System. The proposal is then weighed with the existing system analytically and the best one is selected. The proposal is then presented to the user for an endorsement by the user. The proposal is reviewed on user request and suitable changes are made. This is a loop that ends as soon as the user is satisfied with the proposal.

2.2 STAKEHOLDERS IN THIS PROJECT

2.2.1 ADMIN

The website owner has a wide range of power across Note Sharing Pro. The owner manages the infrastructure of site content (such as templates, style sheets, and processes). Owner privileges include the ability to make major changes to the system, typically an operating system. The owner also manages software programs such as the database management system. This is the first and very important module in the system. Note Sharing Pro cannot be completed without the presence of the owner. The owner single-handedly manages the system. There is only one owner in the system.

No system exists without any kind of problem. These problems are handled by the owner of the system, such as when any user violates the guidelines of the system or when managing disputes. Here, the owner will play the same role. The owner will monitor the activity of each user. The owner has the right to add new content to the site, delete the content of the site, and also update the site content. The owner will approve

or reject the documents uploaded by the users. They can also check the status of virtual coin transactions.

2.2.2 COLLEGE

The college module is designed to oversee and manage the educational resources shared on the platform. College administrators are granted extensive powers to manage and monitor site content, ensuring it aligns with academic standards and guidelines. They are responsible for approving or rejecting documents uploaded by users, maintaining the quality and relevance of the shared resources. The college administrators can also add, update, or delete content on the site, tailoring the platform to meet the institution's needs. They handle disputes and ensure compliance with platform policies, providing a structured and secure environment for note sharing. Additionally, college administrators can track user activity and document usage, generating reports to gain insights into how the platform is being utilized within their institution.

2.2.3 USER

The user module is designed to provide a seamless and engaging experience for students and faculty members. Students can register and log into the platform, gaining access to a personalized dashboard where they can manage their profiles and view their activity history. They can upload various types of documents, including videos and PDFs, which are subject to approval by the college administrators. Faculty members can upload various types of study material, also faculty is free to access any document without a using any coins. Users can also participate in the global doubt section, posting questions, providing answers, and engaging in discussions with peers and experts. The wallet system allows users to earn or purchase virtual coins to access premium content, incentivizing the sharing of high-quality resources. Through the user module, students and faculty members can efficiently share and access educational materials, fostering a collaborative and resource-rich learning environment.

2.3 SOFTWARE REQUIREMENT SPECIFICATION

2.3.1 ADMIN

1. Secure login using username and password.
2. Dashboard display upon login, showcasing system overview.
3. Document verification functionality.
4. Management of user accounts and documents.
5. Monitoring and management of user activity.
6. Coin allocation and management.
7. Feature for responding to global doubts.
8. Logout functionality.

2.3.2 COLLEGE

1. Secure login using username and password
2. Personalized dashboard upon login.
3. Add and manage college departments and courses.
4. Add and manage faculty members.
5. Review and approve or reject user-uploaded documents.
6. Add, update, or delete site content.
7. Track user activity and generate reports.
8. Handle user disputes and guideline violations.
9. Manage virtual coin distribution and monitor transactions.
10. Moderate discussions and ensure content quality.
11. Securely log out of the system.

2.3.3 USER

1. Secure login using username and password.
2. Personalized dashboard upon login.
3. Document upload and sharing capabilities.
4. Access to a variety of document types, including videos, PDFs, etc.
5. Wallet system for accessing content.
6. Global doubt section for asking questions and engaging in discussions.
7. User profile management.
8. Logout functionality.

Table 2.1. Sign off table

Sl. No.	Name & Designation	Date	Accepted (Yes/No)
1	Leema George Assistant Professor Santhigiri College of Computer Sciences		
2	Alan Reji Developer		
3	Aldrin Biju Developer		

2.4 FEASIBILITY STUDY

Feasibility is defined as the practical extent to which a project can be performed successfully. To evaluate feasibility, a feasibility study is performed, which determines whether the solution considered to accomplish the requirements is practical and workable in the software. Information such as resource availability, cost estimation for software development, benefits of the software to the organization after it is developed and cost to be incurred on its maintenance are considered during the feasibility study. The objective of the feasibility study is to establish the reasons for developing the software that is acceptable to users, adaptable to change and conformable to established standards. Various other objectives of feasibility study are listed below.

- To analyze whether the software will meet organizational requirements.
- To determine whether the software can be implemented using the current technology and within the specified budget and schedule.
- To determine whether the software can be integrated with other existing software. When our project guide as well as our client Ms. Leema George told us regarding the mini project and about Word to the Wise for getting the desired product developed, it comes up with rough idea about what all functions the software must perform and which all features are expected from the software.

Referencing to this information, we do a studies and discussions about whether the desired system and its functionality are feasible to develop and the output of this phase is a feasibility study report that should contained adequate comments and

recommendations.

Various types of feasibility that we checked include technical feasibility, operational feasibility, and economic feasibility.

Technical Feasibility

Technical feasibility assesses the current resources (such as hardware and software) and technology, which are required to accomplish user requirements in the software within the allocated time and budget. For this, the software development team ascertains whether the current resources and technology can be upgraded or added in the software to accomplish specified user requirements. Technical feasibility also performs the following tasks.

- Analyses the technical skills and capabilities of the software development team members.
- Determines whether the relevant technology is stable and established.
- Ascertains that the technology chosen for software development has a large number of users so that they can be consulted when problems arise or improvements are required.

From our perspective there are two languages PHP, HTML and database MySQL which are used to develop this web-based applications. PHP is used in the front end and MySQL is used in the back end. The Word to the Wise is web based and thus can be accessed through any browsers. As we are using these latest technologies which are currently trending and used by a number of developers across the globe, we can say that our project is technically feasible.

Operational Feasibility

Operational feasibility assesses the extent to which the required software performs a series of steps to solve business problems and user requirements. This feasibility is dependent on human resources (software development team) and involves visualizing whether the software will operate after it is developed and be operative once it is installed. Operational feasibility also performs the following tasks.

- Determines whether the problems anticipated in user requirements are of high priority.

- Determines whether the solution suggested by the software development team is acceptable.
- Analyses whether users will adapt to a new software.
- Determines whether the organization is satisfied by the alternative solutions proposed by the software development team.

We found that our project will be satisfied for the client since we were discussing every detail about the software with the client at every step. The most important part of operational feasibility study is the input from client. So, the software is built completely according to the requirements of the client. We have used the current industry standards for the software. Hence, we can say that this software is operationally feasible.

Economic Feasibility

Economic feasibility determines whether the required software is capable of generating financial gains for an organization. It involves the cost incurred on the software development team, estimated cost of hardware and software, cost of performing feasibility study, and so on. For this, it is essential to consider expenses made on purchases (such as hardware purchase) and activities required to carry out software development. In addition, it is necessary to consider the benefits that can be achieved by developing the software. Software is said to be economically feasible if it focuses on the issues listed below.

- Cost incurred on software development to produce long-term gains for an organization.
- Cost required to conduct full software investigation (such as requirements elicitation and requirements analysis).
- Cost of hardware, software, development team, and training.

It is estimated that our project is economically feasible as development cost is very minimal since the tools and technologies used are available online. It's a group student project so there are no personnel costs. Development time is well planned and will not affect other operations and activities of the individuals. Once the system has been developed, the companies purchasing the system will be providing with a manual for training purposes. There is no need to purchase new hardware since the existing computers can still be used to implement the new system.

2.5 SOFTWARE DEVELOPMENT LIFECYCLE MODEL

One of the basic notions of the software development process is SDLC models which stand for Software Development Life Cycle models. SDLC – is a continuous process, which starts from the moment, when it's made a decision to launch the project and it ends at the moment of its full remove from the exploitation. Software development lifecycle (SDLC) is a framework that defines the steps involved in the development of software. It covers the detailed plan for building, deploying and maintaining the software. SDLC defines the complete cycle of development i.e. all the tasks involved in gathering a requirement for the maintenance of a Product.

Some of the common SDLC models are Waterfall Model, V-Shaped Model, Prototype Model, Spiral Model, Iterative Incremental Model, Big Bang Model, Agile Model. We used Agile Model for our Project.

Agile Model

Agile Model is a combination of the Iterative and incremental model. This model focuses more on flexibility while developing a product rather than on the requirement. In the agile methodology after every development iteration, the client is able to see the result and understand if he is satisfied with it or he is not. Extreme programming is one of the practical uses of the agile model. The basis of this model consists of short meetings where we can review our project. In Agile, a product is broken into small incremental builds. It is not developed as a complete product in one go. At the end of each sprint, the project guide verifies the product and after his approval, it is finalized. Client feedback is taken for improvement and his suggestions and enhancement are worked on in the next sprint. Testing is done in each sprint to minimize the risk of any failures.

Advantages of Agile Model:

- It allows more flexibility to adapt to the changes.
- The new feature can be added easily.
- Customer satisfaction as the feedback and suggestions are taken at every stage.
- Risks are minimized thanks to the flexible change process

Disadvantages:

- Lack of documentation.
- If a customer is not clear about how exactly they want the product to be, then the project would fail.
- With all the corrections and changes there is possibility that the project will exceed expected time

2.6 HARDWARE AND SOFTWARE REQUIREMENTS

2.6.1 SOFTWARE SPECIFICATION

This project is built upon the latest technology software.

Front end	:	HTML, JavaScript
Development tool	:	PHP
Database	:	MySQL
Web server	:	WAMP server
Operating System	:	Windows 10

2.6.1.1 PHP

PHP is a server-side scripting language designed for web development but also used as a general-purpose programming language. As of January 2013, PHP was installed on more than 240 million websites (39% of those sampled) and 2.1 million web servers. Originally created by Rasmus Lerdorf in 1994, the reference implementation of PHP is now produced by The PHP Group. While PHP originally stood for Personal Home Page, it now stands for PHP: Hypertext Pre-processor, a recursive acronym.

PHP code can be simply mixed with HTML code, or it can be used in combination with various templating engines and web frameworks. PHP code is usually processed by a PHP interpreter, which is usually implemented as a web server's native module or a Common Gateway Interface (CGI) executable. After the PHP code is interpreted and executed, the web server sends resulting output to its client, usually in form of a part of the generated web page - for example, PHP code can generate a web pages.

HTML code, an image, or some other data. PHP has also evolved to include a command line interface (CLI) capability and can be used in standalone graphical applications.

PHP is free software released under the PHP License. PHP has been widely ported and can be deployed on most web servers on almost every operating system and platform, free of charge.

2.6.1.2 MySQL

MySQL is the world's most popular open-source database software, with over 100 million copies of its software downloaded or distributed throughout its history. With its superior speed, reliability, and ease of use, MySQL has become the preferred choice for Web, Web 2.0, SaaS, ISV, Telecom companies and forward-thinking corporate IT Managers because it eliminates the major problems associated with downtime, maintenance and administration for modern, online applications.

Many of the world's largest and fastest-growing organizations use MySQL to save time and money powering their high-volume Web sites, critical business systems, and packaged software — including industry leaders such as Yahoo!, Alcatel-Lucent, Google, Nokia, YouTube, Wikipedia, and Booking.com.

The flagship MySQL offering is MySQL Enterprise, a comprehensive set of production- tested software, proactive monitoring tools, and premium support services available in an affordable annual subscription.

MySQL is a key part of LAMP (Linux, Apache, MySQL, PHP / Perl / Python), the fast-growing open source enterprise software stack. More and more companies are using LAMP as an alternative to expensive proprietary software stacks because of its lower cost and freedom from platform lock-in.

MySQL was originally founded and developed in Sweden by two Swedes and a Finn: David Axmark, Allan Larsson and Michael "Monty" Widenius, who had worked together since the 1980's. MySQL, the most popular Open Source SQL database management system, is developed, distributed, and supported by Oracle Corporation.

MySQL is a database management system. A database is a structured collection of data. It may be anything from a simple shopping list to a picture gallery or the vast amounts of information in a corporate network. To add, access, and process data stored in a computer database, you need a database management system such as MySQL Server. Since computers are very good at handling large amounts of data, database management

systems play a central role in computing, as standalone utilities, or as parts of other applications.

MySQL databases are relational. A relational database stores data in separate tables rather than putting all the data in one big storeroom. The database structures are organized into physical files optimized for speed. The logical model, with objects such as databases, tables, views, rows, and columns, offers a flexible programming environment. You set up rules governing the relationships between different data fields, such as one-to one, one-to- many, unique, required or optional, and —pointers between different tables. The database enforces these rules, so that with a well-designed database, your application never sees inconsistent, duplicate, orphan, out-of-date, or missing data.

The SQL part of —MySQL stands for —Structured Query Language. SQL is the most common standardized language used to access databases. Depending on your programming environment, you might enter SQL directly (for example, to generate reports), embed SQL statements into code written in another language, or use a language- specific API that hides the SQL syntax.

SQL is defined by the ANSI/ISO SQL Standard. The SQL standard has been evolving since 1986 and several versions exist. In this manual, —SQL-92 refers to the standard released in 1992, —SQL:1999 refers to the standard released in 1999, and —SQL:2003 refers to the current version of the standard.

We use the phrase —the SQL standard to mean the current version of the SQL Standard at any time. MySQL software is Open Source. Open Source means that it is possible for anyone to use and modify the software. Anybody can download the MySQL software from the Internet and use it without paying anything. If you wish, you may study the source code and change it to suit your needs.

The MySQL software use GPL (GNU General Public License), <http://www.fsf.org/licenses/>, to define what you may and may not do with the software in different situations. If you feel uncomfortable with the GPL need to embed MySQL code into a commercial application, you can buy a commercially licensed version from us.

The MySQL Database Server is very fast, reliable, scalable, and easy to use. If that is what you are looking for, you should give it a try. MySQL Server can run comfortably on a desktop or laptop, alongside your other applications, web servers, and so on, requiring little or no attention. If you dedicate an entire machine to MySQL, you can adjust the settings to take advantage of all the memory, CPU power, and I/O capacity available. MySQL can also scale up to clusters of machines, networked together.

MySQL Server was originally developed to handle large databases much faster than existing solutions and has been successfully used in highly demanding production environments for several years. Although under constant development, MySQL Server today offers a rich and useful set of functions. Its connectivity, speed, and security make MySQL Server highly suited for accessing databases on the Internet. MySQL Server works in client/server or embedded systems.

The MySQL Database Software is a client/server system that consists of a multi-threaded SQL server that supports different backends, several different client programs and libraries, administrative tools, and a wide range of application programming interfaces (APIs). We also provide MySQL Server as an embedded multi-threaded library that you can link into your application to get a smaller, faster, easier-to-manage standalone product.

A large amount of contributed MySQL software is available. MySQL Server has a practical set of features developed in close cooperation with our users. It is very likely that your favorite application or language supports the MySQL Database Server.

2.6.1.3 WAMP SERVER

WAMP Server is a Windows web development environment. It allows you to create web applications with Apache2, PHP and a MySQL database. Alongside, PhpMyAdmin allows you to manage easily your databases. WAMP Server refers to a software stack for the Microsoft Windows operating system, created by Romain Bourdon and consisting of the Apache web server, Open SSL for SSL support, MySQL database and PHP programming language. WAMP Server is a Web development platform on Windows that allows you to create dynamic Web applications with Apache2, PHP, MySQL and MariaDB. WampServer automatically installs everything you need to intuitively developed Web applications. You will be able to tune your server without even touching its setting files. Best of all, WampServer is available for

free (under GPML license) in both 32- and 64-bit versions. WampServer is not compatible with Windows XP, SP3, or Windows Server 2003.

WAMP Server's functionalities are very complete and easy to use so we won't explain here how to use them.

With a left click on WAMP Server's icon, you will be able to:

- manage your Apache and MySQL services
- switch online/offline (give access to everyone or only localhost)
- install and switch Apache, MySQL and PHP releases
- manage your server's settings
- access your logs
- access your settings files
- create alias

2.6.1.4 VISUAL STUDIO

Visual Studio Code is a source-code editor that can be used with a variety of programming languages, including C, C#, C++, FORTRAN, Go, Java, JavaScript, Node.js, Python, Rust, and Julia. It is based on the Electron framework, which is used to develop Node.js web applications that run on the Blink layout engine. Visual Studio Code employs the same editor component (code named "Monaco") used in Azure DevOps.

Out of the box, Visual Studio Code includes basic support for most common programming languages. This basic support includes syntax highlighting, bracket matching, code folding, and configurable snippets. Visual Studio Code also ships with Intelligent for JavaScript, Type Script, JSON, CSS, and HTML. Support for additional languages can be provided by freely available extensions on the VS Code Marketplace. Instead of a project system, it allows users to open one or more directories, which can then be saved in work spaces for future reuse. This allows it to operate as a language agnostic code editor for any language. It supports many programming languages and a set of features that differs per language. Unwanted files and folders can be excluded from the project tree via the settings. Many Visual Studio Code features are not exposed through menus or the user interface but can be accessed via the command palette.

2.6.1.5 WINDOWS 11

Operating System is defined as a program that manages the computer hardware. An operating system can be viewed as a scheduler, where it has resources for which it has charge. Resources include CPU, memory, I/O device and disk space. In another view, the operating system is a new machine. The third view is that operating system is a multiplexer which allows sharing of resources provides protection from interference and provides a level of cooperation between users. This project is developed using Windows 10 as the operating system and supports its latest versions. Windows 10 is a series of personal computer operating systems produced by Microsoft as part of its Windows NT family of operating systems. It is the successor to Windows 10, and was released to manufacturing on July 15, 2019, and to retail on July 29, 2019. One of Windows 10's most notable features is support for universal apps. Windows 10 also introduced the Microsoft Edge web browser, a virtual desktop system, a window and desktop management feature called Task View, support for fingerprint and face recognition login, new security features for enterprise environments, and DirectX12. Windows 11 received mostly positive reviews upon its original release in July 2019. Critics praised Microsoft's decision to provide a desktop-oriented interfacing line with previous versions of Windows, contrasting the tablet-oriented approach of 8, although Windows 11's touch-oriented user interface mode was criticized for containing regressions upon the touch-oriented interface of Windows 10. Critics also praised the improvements to Windows 11's bundled software over Windows 10, Xbox Live integration, as well as the functionality and capabilities of the Cortana personal assistant and the replacement of Internet Explorer with Microsoft Edge. However, media outlets have been critical of changes to operating system behaviors, including mandatory update installation, privacy concerns over data collection performed by the OS for Microsoft and its partners and the adware-like tactics used to promote the operating system on its release.

2.6.1.6 MICROSOFT WORD

Microsoft Word (or simply Word) is a word processor developed by Microsoft. It was first released on October 25, 1983 under the name Multi-Tool Word for Xenix systems. Subsequent versions were later written for several other platforms including IBM PCs running DOS (1983), Apple Macintosh running the Classic Mac OS (1985), AT&T

Unix PC (1985), Atari ST (1988), OS/2 (1989), Microsoft Windows (1989), SCO Unix (1994), and macOS (formerly OS X; 2001).

Commercial versions of Word are licensed as a standalone product or as a component of Microsoft Office, Windows RT or the discontinued Microsoft Works suite. Unlike most MS-DOS programs at the time, Microsoft Word was designed to be used with a mouse. Advertisements depicted the Microsoft Mouse, and described Word as a WYSIWYG, windowed word processor with the ability to undo and display bold, italic, and underlined text, although it could not render fonts. It was not initially popular, since its user interface was different from the leading word processor at the time, WordStar. However, Microsoft steadily improved the product, releasing versions 2.0 through 5.0 over the next six years. In 1985, Microsoft ported Word to the classic Mac OS (known as Macintosh System Software at the time). This was made easier by Word for DOS having been designed for use with high-resolution displays and laser printers, even though none were yet available to the general public. Following the precedents of Lisa Write and MacWrite, Word for Mac OS added true WYSIWYG features. It fulfilled a need for a word processor that was more capable than MacWrite. After its release, Word for Mac OS's sales were higher than its MS-DOS counterpart for at least four years.

2.6.1.7 SMARTDRAW

SmartDraw is a diagram tool used to make flowcharts, organization charts, mind maps, project charts, and other business visuals. SmartDraw has two versions: an online edition and a downloadable edition for Windows desktop.

SmartDraw integrates with Microsoft Office products including Word, PowerPoint, and Excel and G Suite applications like Google Docs and Google Sheets. SmartDraw has apps for Atlassian's Confluence, Jira, and Trello. SmartDraw is compatible with Google Drive, Dropbox, Box, and OneDrive.

Since 1994, the mission of SmartDraw Software has been to expand the ways in which people communicate so that we can clearly understand each other, make informed decisions, and work together to improve our businesses and the world. We accomplish this by creating software and services that make it possible for people to capture and present information as visuals, while being a pleasure to use. In 2019, we took this to the next level by launching Visual Script, which makes it easy to visualize data in relational formats like trees, flows, and timelines, automatically, without any human

input. Visual Script is a relationship visualization platform that empowers organizations to visualize data across siloed ecosystems and gain critical insights in real-time. Today, SmartDraw Software is one of the most sophisticated digital marketing organizations in the world with over 90,000 unique visitors to our website each business day and in excess of 3,000,000 installations of our apps each year. SmartDraw is used by more than half of the Fortune 500 and by over 250,000 public and private enterprises of all sizes around the world. Privately held, SmartDraw Software is headquartered in San Diego, California.

2.6.2 HARDWARE REQUIREMENTS

The selection of hardware configuring is a very task related to the software development, particularly inefficient RAM may affect adversely on the speed and corresponding on the efficiency of the entire system. The processor should be powerful to handle all the operations. The hard disk should have the sufficient to solve the database and the application.

Hardware used for development:

CPU	: Intel i5 Processor
Memory	: 4 GB
Cache	: 6 MB
Hard Disk	: 1 TB
Monitor	: 15.6" Monitor
Keyboard	: Standard 108 keys Enhanced Keyboard
Mouse	: Optical Mouse

Minimum Hardware Required for Implementation:

CPU	: Pentium IV Processor
Memory	: 256MB Above
Cache	: 512 KB Above
Hard Disk	: 20 GB Above

Monitor : Any

Keyboard : Any

Mouse : Any

3. SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

A system architecture or system's architecture is the conceptual model that defines the structure, behavior, and more views of a system. An architecture description is a formal description and representation of a system, organized in a way that supports reasoning about the structures of the system,

System architecture can comprise system components, the externally visible properties of those components, the relationships (e.g. the behavior) between them. It can provide a plan from which products can be procured, and systems developed, that will work together to implement the overall system. There have been efforts to formalize languages to describe system architecture; collectively these are called architecture description languages (ADLs).

The system architecture can best be thought of as a set of representations of an existing (or to be created) system. It is used to convey the informational content of the elements comprising a system, the relationships among those elements, and the rules governing those relationships. The architectural components and set of relationships between these components that architecture describes may consist of hardware, software, documentation, facilities, manual procedures, or roles played by organizations or people. System architecture is primarily concerned with the internal interfaces among the system's components or subsystems, and the interface between the system and its external environment, especially the user.

The structural design reduces complexity, facilitates change and result in easier implementation by encouraging parallel development of different parts of the system. The procedural design transforms structural elements of program architecture into a procedural description of software components. The architectural design considers architecture as the most important functional requirement. The system is based on the three-tier architecture.

The first level is the user interface (presentation logic), which displays controls, receives and validates user input. The second level is the business layer (business logic) where the application specific logic takes place. The third level is the data layer where the application information is stored in files or database. It contains logic about to

retrieve and update data. The important feature about the three-tier design is that information only travels from one level to an adjacent level.

3.2 MODULE DESIGN

Modular programming is a software design technique that emphasizes separating the functionality of a program into independent, interchangeable modules, such that each contains everything necessary to execute only one aspect of the desired functionality. Conceptually, modules represent a separation of concerns, and improve maintainability by enforcing logical boundaries between components.

Different modules of this project include.

1. User Authentication

This module enables both administrators and users to securely log into Note Sharing Pro. Administrators log in using their unique username and password, which grants them access to all system functions and management tools. College log in using their unique username and password, which grants them access to all system functions and management tools allotted for modification of their college side. Users also log in with their credentials to access and interact with content. This Module supports password changes and resets for both admins and users, ensuring they can maintain account security and recover access if needed.

2. Document Management

This module oversees the management of documents shared on the platform. Users can upload various types of documents, question paper and PDFs, which are then subject to approval by college. College has the authority to review, approve, or reject document submissions to ensure quality and relevance. This module also includes features for organizing documents into categories and managing access permissions based on document types and user roles. Effective document management ensures that all shared content meets the platform's standards and is accessible to users as intended.

3. Wallet System

This module manages the virtual currency used for accessing content on Note Sharing Pro. New users receive an allocation of free virtual coins, which they can use to access documents or purchase additional coins. Users can view their current coin balance, make purchases through various online payment methods, and spend coins to access or download content. Administrators and college oversee coin transactions, manage allocations, and monitor usage to ensure fairness and prevent abuse. This system facilitates a flexible and engaging way for users to interact with content while providing a revenue model for the platform.

4. Global Doubt Session

This module supports community engagement and knowledge exchange through the global doubt section. Users can post questions, seek clarifications, and participate in discussions on various topics. Faculty and other users can provide answers and contribute to discussions, fostering a collaborative learning environment. This section is moderated to ensure that discussions remain productive and relevant, with features for reporting inappropriate content or behavior. The global doubt section enhances the platform's functionality by allowing users to share expertise and solve problems collectively.

5. Generating Report

This module allows administrators to generate and review detailed reports on platform activity. Reports include metrics on document submissions, user interactions, and coin transactions. The system supports various report formats, such as pie charts, Excel spreadsheets, and PDFs, providing insights into usage patterns and content popularity. Administrators can generate reports on user activity, document access statistics, and financial transactions, which helps in managing the platform effectively and making informed decisions. The reporting module plays a crucial role in monitoring system performance and optimizing user engagement.

3.3 DATABASE DESIGN

A database is a collection of interrelated data stored with minimum redundancy to serve many users quickly and efficiently. The general objective is to make information access easy, quick, inexpensive and flexible for the users. The general theme behind a database is to integrate all information. Database design is recognized as a standard of management information system and is available virtually for every computer system.

In database design several specific objectives are considered:

- Ease of learning and use
- Controlled redundancy
- Data independence
- More information at low cost
- Accuracy and integrity
- Recovery from failure
- Privacy and security
- Performance

3.3.1 Normalization

Designing a database is complete task and the normalization theory is a useful aid in the design process. The process of normalization is concerned with transformation of conceptual schema into computer representation form. There will be need for most databases to grow by adding new attributes and new relations. The data will be used in new ways. Tuples will be added and deleted. Information stored may undergo updating also. New association may also be added. In such situations the performance of a database is entirely depend upon its design. A bad database design may lead to certain undesirable things like:

- Repetition of information
- Inability to represent certain information
- Loss of information

To minimize these anomalies, Normalization may be used. If the database is in a normalized form, the data can be growing without, in most cases, forcing the rewriting application programs. This is important because of the excessive and growing cost of maintaining an organization's application programs and its data from the disrupting

effects of database growth. As the quality of application programs increases, the cost of maintaining the without normalization will rise to prohibitive levels. A normalized database can also encompass many related activities of an organization thereby minimizing the need for rewriting the applications of programs. Thus, normalization helps one attain a good database design and there by ensures continued efficiency of database.

Normalization theory is built around the concept of normal forms. A relation is said to be in normal form if it satisfies a certain specified set of constraints. For example, a relation is said to be in first normal form (1NF) if it satisfies the constraint that it contains atomic values only. Thus, every normalized relation is in 1NF. Numerous normal forms have been defined. Codd defined the first three normal forms.

All normalized relations are in 1NF, some 1NF relations are also in 2NF and some 2NF relations are also in 3NF. 2NF relations are more desirable than 1NF and 3NF are more desirable than 2NF. That is, the database designer should prefer 3NF than 1NF or 2NF. Normalization procedure states that a relation that is in some given normal form can be converted into a set of relations in a more desirable form. We can define this procedure as the successive reduction of a given collection of relations to some more desirable form. This procedure is reversible. That is, it is always possible to take the output from the procedure and convert them back into input. In this process, no information is lost. So, it is also called “no loss decomposition”.

First Normal Form

A relation is in first normal form (1NF) if and all its attributes are based on single domain. The objective of normalizing a table is to remove its repeating groups and ensure that all entries of the resulting table have at most single value.

Second Normal Form

A table is said to be second Normal Form (2NF), when it is in 1NF and every attribute in record is functionally dependent upon the whole key, and not just a part of the key.

Third Normal Form

A table is in third Normal Form (3NF), when it is in 2NF and every non-key attribute is functionally dependent on just the primary key.

3.3.2 Table Structure

Table is a collection of complete details about a particular subject. These data are saved in rows and Columns. The data of each Row are different units. Hence, rows are called RECORDS and Columns of each row are called FIELDS.

Data is stored in tables, which is available in the backend the items and data, which are entered in the input, form id directly stored in this table using linking of database. We can link more than one table to input forms. We can collect the details from the different tables to display on the output.

There are mainly 11 tables in the project. They are,

1. tbl_adminlogin
2. tbl_district
3. tbl_location
4. tbl_department
5. tbl_course
6. tbl_college
7. tbl_collegedepartment
8. tbl_collegecourse
9. tbl_faculty
10. tbl_student
11. tbl_subject
12. tbl_document
13. tbl_wallet
14. tbl_access

1. Table: tbl_adminlogin

Description: This table is used to store admin login details.

Table 3.1 tbl_adminlogin

Field Name	Data Type	Constraint	Description of Field
login_id	Int	Primary Key	This field is used to uniquely identify login id of admin, and the values are automatically generated.
username	Varchar(20)	Not Null	This is field is used to store the username of the admin
password	Varchar(30)		This is field is used to store the password of the admin

2. Table: tbl_district

Description: This table is used to store the district details.

Table 3.2 tbl_district

Field Name	Data Type	Constraint	Description of Field
district_id	Int	Primary Key	This field is used to uniquely identify district, and the values are automatically generated.
district_name	Varchar(20)	Not Null	This field stores the district name.

3. Table: tbl_location

Description : This table is used to store the department information.

Table 3.3 tbl_location

Field Name	Data Type	Constraint	Description of Field
location_id	Int	Primary Key	This field is used to uniquely identify location, and the values are automatically generated.
district_id	Int	Foreign key	This field stores the district id of the location
location_name	Varchar(20)	Not Null	This field stores the name of the location under the district.

4. Table: tbl_department

Description : This table is used to store department details.

Table 3.4 tbl_department

Field Name	Data Type	Constraint	Description of Field
department_id	Int	Primary Key	This field is used to uniquely identify department, and the values are automatically generated.
Department_name	Varchar (20)	Not Null	This field stores the name of the department.
Department_logo	Varchar(100)	Not Null	This field stores the image of the department.

5. Table: tbl_course

Description : This table is used to store the course details.

Table 3.5 tbl_course

Field Name	Data Type	Constraint	Description of Field
course_id	Int	Primary Key	This field is used to uniquely identify courses, and the values are automatically generated.
course_name	Varchar(20)	Not Null	This field stores the name of the courses.
sem_count	Int	Not Null	This field stores the number of the semesters in each course.
Department_id	Int	Foreign key	This field stores the department id of the course

6. Table: tbl_college

Description : This table is used to store the college details.

Table 3.6 tbl_college

Field Name	Data Type	Constraint	Description of Field
College_id	Int	Primary Key	This field is used to uniquely identify college, and the values are automatically generated.
College_name	Varchar(100)	Not Null	This field stores the name of the college.
District_id	Int	Foreign key	This field stores the district id of the college
Location_id	Int	Foreign key	This field stores the location id of the college
College_phone_number	Bigint	Not null	This field stores the phone number of the college
College_email	Varchar(30)	Not null	This field stores the email of the college
Reg_date	Date	Not null	This field stores the registration date of the college.
Image	Varchar(100)	Not Null	This is field is used to store the image of the college.
username	Varchar(20)	Not null	This field stores the username of the college.
Password	Varchar (30)	Not null	This field stores the password of the college.

7. Table: **tbl_college_department**

Description : This table is used to store the department details in a college.

Table 3.7 tbl_college_department

Field Name	Data Type	Constraint	Description of Field
College_department_id	Int	Primary Key	This field is used to uniquely identify department in a college , and the values are automatically generated.
Department_id	Int	Foreign key	This field stores the department id of the department.
College_id	Int	Foreign key	This field stores the college id of the college.

8. Table: **tbl_college_course**

Description : This table is used to store the course details in a college.

Table 3.8 tbl college_department

Field Name	Data Type	Constraint	Description of Field
College_course_id	Int	Primary Key	This field is used to uniquely identify course in a college, and the values are automatically generated.
Course_id	Int	Foreign key	This field stores the course id of the course.
College_Department_id	Int	Foreign key	This field stores the college department id of the college department.

9. Table: **tbl_faculty**

Description : This table is used to store the faculty details.

Table 3.9 tbl_faculty

Field Name	Data Type	Constraint	Description of Field
faculty_id	Int	Primary Key	This field is used to uniquely identify faculty, and the values are automatically generated.
College_id	int	Foreign key	This field stores the college id of the college
name	Varchar(20)	Not Null	This field stores the name of the faculty.
Phone	Varchar(20)	Not Null	This is field stores the phone number of the faculty.
College_department_id	int	Foreign key	This field stores the college department id of the college department
Email	Varchar(30)	Not Null	This is field stores the email of the faculty.
email	Varchar(20)	Not Null	This field stores the email of the faculty.
regdate	date	Not null	This field stores the registration date of the faculty
Username	Varchar(20)	Not Null	This field stores the username of the faculty.
Password	Varchar(20)	Not Null	This field stores the password of the faculty.

10. Table: tbl_subjects

Description : This table is used to store the subject details in a course.

Table 3.10 tbl_subject

Field Name	Data Type	Constraint	Description of Field
Subject_id	Int	Primary Key	This field is used to uniquely identify subject in a course, and the values are automatically generated.
Subject_name	Varchar(20)	Not null	This field stores the name of the subject .
Semester	Int	Not null	This field stores the semesters of subject.
College_course_id	Foreign key	Not null	This field stores the college course id of the college course.

11. Table: tbl_student

Description : This table is used to store the student details.

Table 3.11 tbl_student

Field Name	Data Type	Constraint	Description of Field
student_id	Int	Primary Key	This field is used to uniquely identify user, and the values are automatically generated.
College_id	int	Foreign key	This field stores the college id of the collection.
Name	Varchar(20)	Not null	This field stores the name of the user.
Phone	Varchar(20)	Not Null	This is field stores the phone number of the student.
Email	Varchar(30)	Not Null	This is field stores the email of the student.
College_department_id	int	Foreign key	This field stores the college department id of the college department
College_course_id	int	Foreign key	This field stores the college course id of the college course
District_id	int	Foreign key	This field stores the district id of the district
Location_id	int	Foreign key	This field stores the location id of the location
Username	Varchar(20)	Not Null	This field stores the email of the user.
User_Password	Varchar(20)	Not Null	This field stores the password of the user.
Wallet_id	int	Foreign key	This field stores the wallet id of the wallet
regdate	date	Not null	This field stores the date of the student registration

12. Table: tbl_document

Description : This table is used to store the details about document.

Table 3.12 tbl_document

Field Name	Data Type	Constraint	Description of Field
document_id	Int	Primary Key	This field is used to uniquely identify type of user, and the values are automatically generated.
name	Varchar (20)	Not null	This is field is used to store the document of the user
category	Varchar (20)	Not null	This is field is used to store the category of the document
Subject_id	Int	Foreign key	This field stores the subject id of the subject
documentfilename	Varchar (100)	Not null	This field stores the document file name of the document
status	Varchar(20)	Not null	This field stores the status of the document
remark	Varchar(100)	Null	This field stores the remark of document
coin	int	Not null	This field stores the coin allocated
upload_id	int	Foreign key	This field stores the id of the user uploaded the document

13. Table: tbl_walletinfo

Description : This table is used to store the wallet details.

Table 3.13 tbl_walletinfo

Field Name	Data Type	Constraint	Description of Field
Wallet_id	Int	Primary Key	This field is used to uniquely identify wallet, and the values are automatically generated.
Wallet_no	Int	Foreign Key	This field stores the wallet id of the user.
Coin	Int	Not Null	This field stores the coin count of the wallet.

14. Table: tbl_access

Description: This table is used to store the access information about the document.

Table 3.14 tbl_usertype

Field Name	Data Type	Constraint	Description of Field
access_id	Int	Primary Key	This field is used to uniquely identify type of user, and the values are automatically generated.
Document_id	int	Foreign key	This field stores the document id of the document
Access_date	date	Not null	This field stores the access of document

3.3.3 Data Flow Diagram

3.3.3.1 Introduction to Data Flow Diagrams

Data Flow Diagram is a network that describes the flow of data and processes that change, or transform, data throughout the system. This network is constructed by use a set of symbols that do not imply a physical implementation. It is a graphical tool for structured analysis of the system requirements. DFD models a system by using external entities from which data flows to a process, which transforms the data and creates, output-data-flows which go to other processes or external entities or files. Data in files may also flow to processes as inputs.

There are various symbols used in a DFD. Bubbles represent the processes. Named arrows indicate the data flow. External entities are represented by rectangles. Entities supplying data are known as sources and those that consume data are called sinks. Data are stored in a data store by a process in the system. Each component in a DFD is labelled with a descriptive name. Process names are further identified with a number.

The Data Flow Diagram shows the logical flow of a system and defines the boundaries of the system. For a candidate system, it describes the input (source), outputs (destination), database (files) and procedures (data flow), all in a format that meet the user's requirements.

The main merit of DFD is that it can provide an overview of system requirements, what data a system would process, what transformations of data are done, what files are used, and where the results flow.

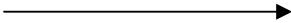
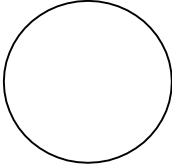
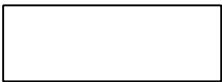
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Rules for constructing a Data Flow Diagram

1. Arrows should not cross each other
2. Squares, circles and files must bear names.
3. Decomposed data flow squares and circles can have same time
4. Choose meaningful names for data flow
5. Draw all data flows around the outside of the diagram

Basic Data Flow Diagram Symbols

Table 3.12 Data Flow Diagram Symbols

	A data flow is a route, which enables packets of data to travel from one point to another. Data may flow from a source to a process and from data store or process. An arrow line depicts the flow, with arrow head pointing in the direction of the flow.
	Circles stands for process that converts data in to information. A process represents transformation where incoming data flows are changed into outgoing data flows.
	A data store is a repository of data that is to be stored for use by a one or more process may be as simple as buffer or queue or sophisticated as relational database. They should have clear names. If a process merely uses the content of store and does not alter it, the arrowhead goes only from the store to the process. If a process alters the details in the store, then a double-headed arrow is used.
	A source or sink is a person or part of an organization, which enters or receives information from the system, but is considered to be outside the contest of data flow model.

3.3.3.2 Data Flow Diagram

Each component in a DFD is labelled with a descriptive name. Process name is further identified with number. Context level DFD is draw first. Then the process is decomposed into several elementary levels and is represented in the order of importance. A DFD describes what data flow (logical) rather than how they are processed, so it does not depend on hardware, software, and data structure or file organization. A DFD methodology is quite effective; especially when the required design.



Fig 3.1 Zeroth level DFD for Note Sharing Pro

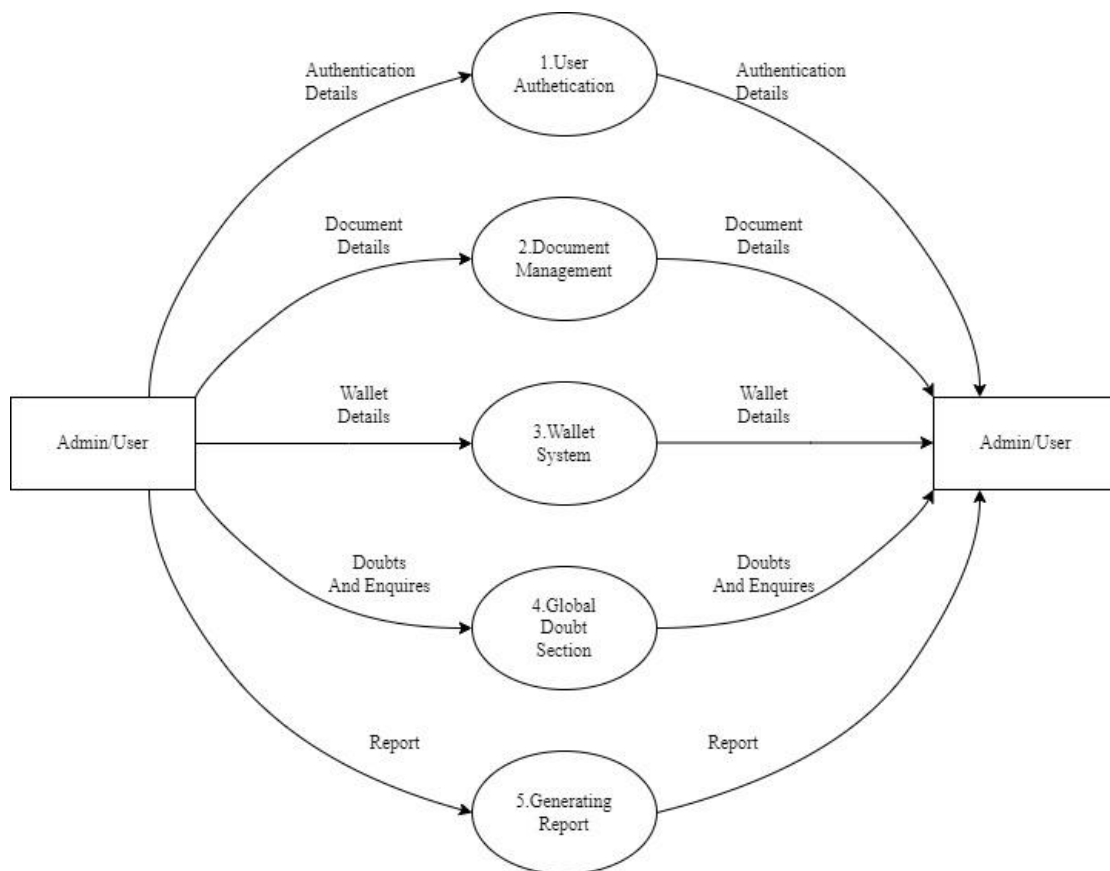


Fig 3.2 First Level DFD for Note Sharing Pro

